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# DEEFAKE DETECTION with ResNeXt-50 and LSTMs

## ABSTRACT

A.I. has grown to epidemic proportions over the last years as its applied in almost all sectors to allocate workload from humans but end up being done effectively with no human intervention. A branch of A.I. called deep learning, which operates by mimicking human judgment and action through neural network systems. Nonetheless, with the increase height of the two platforms have been experienced sufficient cases of misguided individuals using tools to recycle videos, audios, and texts to achieve their agendas. This insinuates a due assumption that Generative Adversarial Networks, GANs, are central to the development of believable deepfakes. GANs have developed a crucial ability to generate videos that replace frames with material from another video source to create deepfakes videos. While GANs serves various purposes such as entertainment, teaching, and experimentation, malicious actors can misuse these deep learning techniques to manipulate videos, impacting the privacy of individuals in society. This paper conducts an analysis of different deepfake detection models, comparing their efficacy and discussing potential future extensions of deepfake technology. The research introduces a deepfake detection model established on Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN). The framework employs ResNext50 for feature extraction at the frame level, and LSTM (Long ShortTerm Memory) is utilized to classify videos based on these frame-level features. The study utilizes diverse datasets such as deepfake detection challenge dataset (DFDC) and Face Forensics deepfake collections (FF++), combining them to achieve a high-accuracy model capable of accurately discerning between real and deepfake videos. The findings of this research contribute to the ongoing efforts to enhance deepfake detection capabilities and safeguard privacy in an era increasingly shaped by artificial intelligence.

**Keywords:-** Artificial Intelligence, Generative Adversarial Networks (GANs), Deep Learning, ResNext, LSTM.

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## ACRONYMS

- Deepfakes (DF)
- ResNext (RN)
- Long Short Term Memory (LSTM)
- Convolutional Neural Network (CNN)
- Recurrent Neural Networks (RNN)
- Machine Learning (ML)
- Deep Learning (DL)
- Artificial Intelligence (AI)
- Department of Computer Science & Engineering (CSE)
- Bachelor of Technology (B.Tech)
- Assistant Professor (Asst. Prof.)
- Jan-May 2024 (J-M 2024)
- Turnitin (TII)
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## NOMENCLATURE

- **DF:** Deepfakes
- **RN:** ResNext
- **LSTM:** Long Short Term Memory
- **CNN:** Convolutional Neural Network
- **RNN:** Recurrent Neural Networks
- **DL:** Deep Learning
- **ML:** Machine Learning
- **AI:** Artificial Intelligence
- **CSE:** Computer Science & Engineering
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## CHAPTER 1: INTRODUCTION

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In today's world, where smart devices are ever-present and social media is extensively used to capture and share moments from daily life, the rise of advanced AI technologies brings significant concerns about media manipulation. Despite these concerns, AI offers numerous advantages, boosting productivity and capacity without tiring. AI finds applications across many fields, including entertainment, education, and forecasting future events such as stock market trends, weather changes, and health conditions in both people and plants. AI's integration into social media platforms enhances user experiences, while online third-party AI tools use the technology for manipulative purposes, creating fake news and misleading information through modified videos, audio, and text. Documented instances exist of AI producing deepfake videos and fake voices, spreading content fueled by conspiracy theories and privacy invasions. Academic research has shown cases where AI alters the voices of famous celebrities for entertaining memes on social media platforms. Moreover, companies use AI to create deepfake videos of celebrities endorsing products, which not only poses privacy risks but also spreads misinformation. Given the increasing capabilities of AI, it's essential to develop methods to detect and prevent media manipulation, thereby safeguarding privacy and fighting the spread of false information. The global issue of deepfake manipulation allows malicious actors to distort videos or audio for personal gain, leading to the extensive distribution of such content online. This invasion of individuals' personal lives can significantly damage the reputations of celebrities and other public figures.

Our project addresses this critical need by proposing a novel deep learning-based approach for detecting deep fakes. By leveraging advanced machine learning algorithms, we aim to develop a system capable of accurately identifying manipulated videos, thus safeguarding the integrity of online content. Through this endeavor, we hope to contribute to the ongoing efforts to combat the spread of misinformation and uphold the principles of truth and authenticity in digital media.

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## CHAPTER 2: LITERATURE SURVEY

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Researchers are actively studying various techniques to detect deepfakes using deep learning architectures. One common approach focuses on facial data, as deepfake methods typically introduce unique artifacts and inconsistencies in facial regions. By concentrating on facial features, these methods aim to identify the signs of manipulation. Researchers are employing various deep learning methods for this purpose, including CNN-based, RNN-based, and other specialized architectures like ResNext50 or LSTM, particularly focusing on facial datasets.

**CNN-based Methods:** Convolutional Neural Networks (CNNs) are effective for detecting deepfakes due to their strengths in image analysis and feature extraction. For example, Dang et al. (2020) utilized CNN architectures to detect inconsistencies and alterations in facial features, demonstrating their approach's effectiveness in differentiating between real and manipulated content [3].

**RNN-based Methods:** Recurrent Neural Networks (RNNs) are able to effectively identify patterns over time and anomalies within video sequences, making them suitable for deepfake detection tasks. Li et al. (2019) contributed to this area with their paper "Detecting Face-Swapped Videos Using Recurrent Neural Networks," employing RNN techniques to analyze temporal patterns in facial alterations, highlighting their utility in identifying deepfake videos [4].

**Specific Architectures on Face-Only Datasets:** Researchers have also explored specialized architectures like ResNext50 and LSTM for deepfake detection on facial datasets. Dolhansky et al. (2020) used the ResNext50 architecture on facial data in their work "Deepfake Detection Using Convolutional Neural Networks and ResNext50," showcasing the approach's potential in identifying manipulated facial content [5]. Similarly, Truong et al. (2021) applied LSTM networks on facial datasets in their study "Deepfake Detection Using LSTM on Face-Only Data," demonstrating the technique's effectiveness in distinguishing deepfake videos [6].

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## CHAPTER 3: PROBLEM DEFINITION AND SCOPE

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### 3.1 Problem Statement

In recent years, researchers have pursued innovative methodologies, such as employing LSTM-based artificial neural networks for deep fake detection. These systems meticulously analyze digital media, scouring for anomalies and patterns indicative of manipulation, thereby aiding in the identification of authentic content amidst the inundation of deep fakes. The detection of these digitally manipulated creations has become increasingly daunting owing to their heightened realism and widespread accessibility. Exploited for nefarious purposes, deep fakes have been implicated in a range of malicious activities, from stoking political discord to fabricating events and disseminating sensitive material such as revenge porn and extortion attempts. Consequently, there is an urgent imperative to devise robust detection techniques to mitigate the proliferation of deep fakes across social media platforms. In recent years, the advancement of deep learning has significantly bolstered the authenticity of manipulated digital content, ushering in the era of AI-generated media, commonly known as deep fakes.

#### 3.1.1 Goals and objectives

Objectives:

- i. Face-Only Representation: Transform videos into face-only representations to concentrate analysis on facial details vital for deepfake detection.
- ii. Feature Extraction with ResNext50: Employ ResNext50, a powerful Convolutional Neural Network (CNN), for feature extraction at the frame level, focusing specifically on facial regions in face-only data.
- iii. Design and implement a user-friendly, robust deepfake detection tool specifically tailored to analyses facial regions in videos, identify and flag manipulated content within face-only data.

### 3.2 Statement of scope

The Deepfake Detection Project aims to confront the urgent challenge of identifying manipulated media content, particularly in light of the increasing sophistication of deepfake technology. At its core, the project seeks to develop a robust deep learning model capable of accurately detecting deepfake images and videos. This detection system is intended to function

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in real-time or near-real-time scenarios, facilitating prompt identification and mitigation of deepfake content across diverse platforms and domains. The project's main deliverables encompass a trained deep learning model, along with a software application or API designed for seamless integration into existing platforms. Additionally, comprehensive documentation and training materials will be provided to assist end-users. The project scope covers various stages, including data collection, model development, evaluation, integration, user interface design, documentation preparation, and training/support provision.

Throughout the project lifecycle, constraints such as time limitations, resource availability, and ethical considerations will be carefully managed. Assumptions regarding access to labeled datasets, expertise in deep learning, and stakeholder commitment are made to facilitate project planning and execution. Recognizing potential risks, such as technical challenges, data quality issues, and adoption concerns, proactive monitoring and mitigation strategies will be implemented. Stakeholders involved in the project include the project team, end-users, and management, each contributing essential roles to ensure the success and impact of the deepfake detection system. This scope statement offers a comprehensive outline of the project's objectives, deliverables, constraints, assumptions, risks, and stakeholders, serving as a guiding framework for its effective implementation.

### 3.2.1 Analysis

In our endeavor to achieve deep feature visualization for enhanced deepfake detection using face-only data, we embark on a comprehensive approach that harnesses the power of multiple datasets and cutting-edge deep learning techniques. Our journey begins with the amalgamation of three renowned datasets: FF++, Celeb-DF, and DFDC. By harmonizing these datasets, we extract crucial labels that serve as the ground truth, providing invaluable insights into the authenticity of each video. However, our focus extends beyond mere label extraction. We recognize the significance of transforming these videos into face-only representations, a critical step that aligns with our deep feature visualization approach. By isolating the facial regions, we can concentrate our analysis on the intricate details and nuances that often hold the key to distinguishing genuine content from manipulated deepfakes. Central to our methodology is the employment of Generative Adversarial Networks (GANs), a powerful technique of deep learning that has revolutionized the field of image synthesis and manipulation. We begin by segmenting the videos into individual frames, allowing for a granular examination of each moment. Subsequently, we substitute the input images for each frame, a process that serves as a precursor to the comprehensive reconstruction of the entire video. This reconstruction phase

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may involve the utilization of autoencoders, ensuring a nuanced understanding of the video's compositional elements. Inspired by a novel deep learning-based approach detailed in existing literature, we capitalize on the identifiable traits inherent to deepfake content. Notably, we synthesize face images of fixed dimensions, a process that often yields discernible artifacts stemming from affine warping. These artifacts serve as critical indicators, guiding our detection approach and enhancing its accuracy. At the core of our approach lies the strategic fusion of two powerful deep learning architectures: a ResNext Convolutional Neural Network (CNN) and a Recurrent Neural Network (RNN) with Long Short-Term Memory (LSTM). This union harnesses the strengths of both architectures, enabling us to effectively capture the temporal inconsistencies introduced by GANs during the reconstruction process. The ResNext50 CNN model, in particular, undergoes rigorous training to directly simulate resolution inconsistencies within affine face wrappings, further refining its ability to detect subtle anomalies. we identified certain key parameters, including: [provide a list of the parameters].

Eye blinking frequency

Enhancement of teeth appearance

Increased distance between the eyes

Addition of moustaches

Duplication of edges for eyes, ears, and nose

Segmentation of iris

Introduction of wrinkles on the face

Inconsistencies in head pose

Variations in face angle

Diversity in skin tone

Facial expressions discrepancies

Differences in lighting conditions

Varied poses of the face

Presence of double chins

Diversity in hairstyle

### 3.2.2 Preprocessing: Focusing on Faces and Frame Selection

To optimize our model for real-time processing on resource-constrained environments, we implement a targeted preprocessing pipeline:

1. Video Splitting and Face Detection: We begin by splitting each video into individual frames. Then, a robust face detection algorithm locates facial regions within each frame, allowing us to focus solely on the most informative areas for deepfake detection and reducing computational overhead compared to processing entire video frames.

2. Frame Cropping and Thresholding: The frames containing detected faces are cropped, retaining only the relevant facial information. To maintain consistency in the model's input and reduce processing requirements, we establish a threshold value ( $p$ ) based on the average frame count across the dataset using the formula:

$$p = \text{mean}(\{k_j\} \mid j = 1 \text{ to } a)$$

Where:

$p$  is the threshold value for the number of frames.

$k_j$  is the number of frames in video  $j$ .

$a$  is the total number of videos in the dataset.

This threshold ensures a uniform number of frames for each processed video.

#### 3. Frame Selection for LSTM Integration:

To analyze data that follows a sequence, we employ networks known as Long Short-Term Memory (LSTM) due to their proficiency in recognizing and remembering patterns over time intervals. Therefore, we select a specific number of consecutive 100 frames with 2048 dimensions from each video, considering our computational resource limitations. This selection enables the LSTM to effectively capture the temporal dynamics of facial expressions and movements within the chosen sequence.

By implementing these preprocessing steps, we create a dataset tailored for real-time deepfake detection, balancing accuracy with unconventional efficiency.

### 3.2.3 Proposed solution

To achieve real-time deepfake detection, finding the right balance between accuracy and computational efficiency is crucial. Our proposed solution focuses on carefully designing a strategy for data acquisition and preprocessing. After conducting thorough research and analysis, we developed the system architecture outlined in our documentation. This architecture establishes the baseline model, specifying the composition of different layers and their quantities. For implementation, we opted to use the PyTorch framework alongside Python 3. PyTorch was chosen for its robust support for CUDA, enabling efficient utilization of Graphic Processing Units (GPUs), and its high level of customizability. To validate the effectiveness of our approach, we evaluated the model using a substantial real-time dataset, including YouTube videos. We utilized the Confusion Matrix approach to assess the accuracy of the trained model.

### 3.3 Methodology

Detecting deepfakes using ResNeXt-50 and LSTM with face-only data involves several key steps. Firstly, we collect a diverse dataset comprising both authentic and deepfake videos featuring human faces. We employ preprocessing techniques like normalization and augmentation to ensure consistency and diversity within the dataset. Subsequently, we utilize the ResNeXt-50 architecture to extract high-level features from the preprocessed face images. This model is fine-tuned on the face-only dataset to optimize its performance for deepfake detection. From the ResNeXt-50 model's output, we extract feature vectors representing the learned representations of each face image. To capture temporal dynamics, we employ LSTM networks. Sequences of feature vectors, corresponding to consecutive frames, are fed into the LSTM network. The LSTM model learns temporal dependencies and patterns within authentic and deepfake face sequences, enabling it to discern manipulated content from genuine footage. We divide the dataset into training, validation, and testing sets for model training and evaluation. The combined ResNeXt-50 and LSTM model is trained on the training set, with parameters optimized to minimize detection error. Validation prevents overfitting, and evaluation assesses the model's generalization ability and performance using metrics such as accuracy, precision, recall, and F1-score. Once trained and evaluated, the deepfake detection model is deployed as either a standalone system or integrated into existing platforms. A user-friendly interface facilitates interaction, allowing users to input video files or stream data in real-time.

Efficient inference algorithms ensure fast and scalable deepfake detection performance, with continuous monitoring and updating to adapt to evolving deepfake generation techniques.

Throughout the process, ethical considerations such as privacy, consent, and bias mitigation are prioritized. Mechanisms are implemented to mitigate biases in both training data and model predictions, ensuring fair and unbiased deepfake detection across diverse demographic groups. By following this methodology, the deepfake detection system effectively leverages the capabilities of ResNeXt-50 and LSTM architectures on face only dataset to combat the proliferation of misinformation and deception in digital media.

### 3.4 Resources Required

#### 3.4.1 Hardware Resources Required

This project necessitates a computer with ample processing capabilities. The high demand for processing power arises from the batch processing of images and videos.

Table 1: Hardware Requirements

| Sr. No. | Parameter          | Minimum Requirement       |
|---------|--------------------|---------------------------|
| 1       | Intel Xeon E5 2637 | 3.5 GHz                   |
| 2       | RAM                | 8 GB                      |
| 3       | Hard Disk          | 150 GB                    |
| 4       | Graphic card       | NVIDIA GeForce (4 GB RAM) |

#### 3.4.2 Software Resources Required

Table 2: Software Requirements

| Sr. No. | Parameter            | Requirement              |
|---------|----------------------|--------------------------|
| 1       | Operating System     | Windows 7+               |
| 2       | Programming Language | Python 3.0               |
| 3       | Framework            | PyTorch 1.4              |
| 4       | Libraries            | OpenCV, Face-recognition |

## CHAPTER 4: SYSTEM ARCHITECTURE

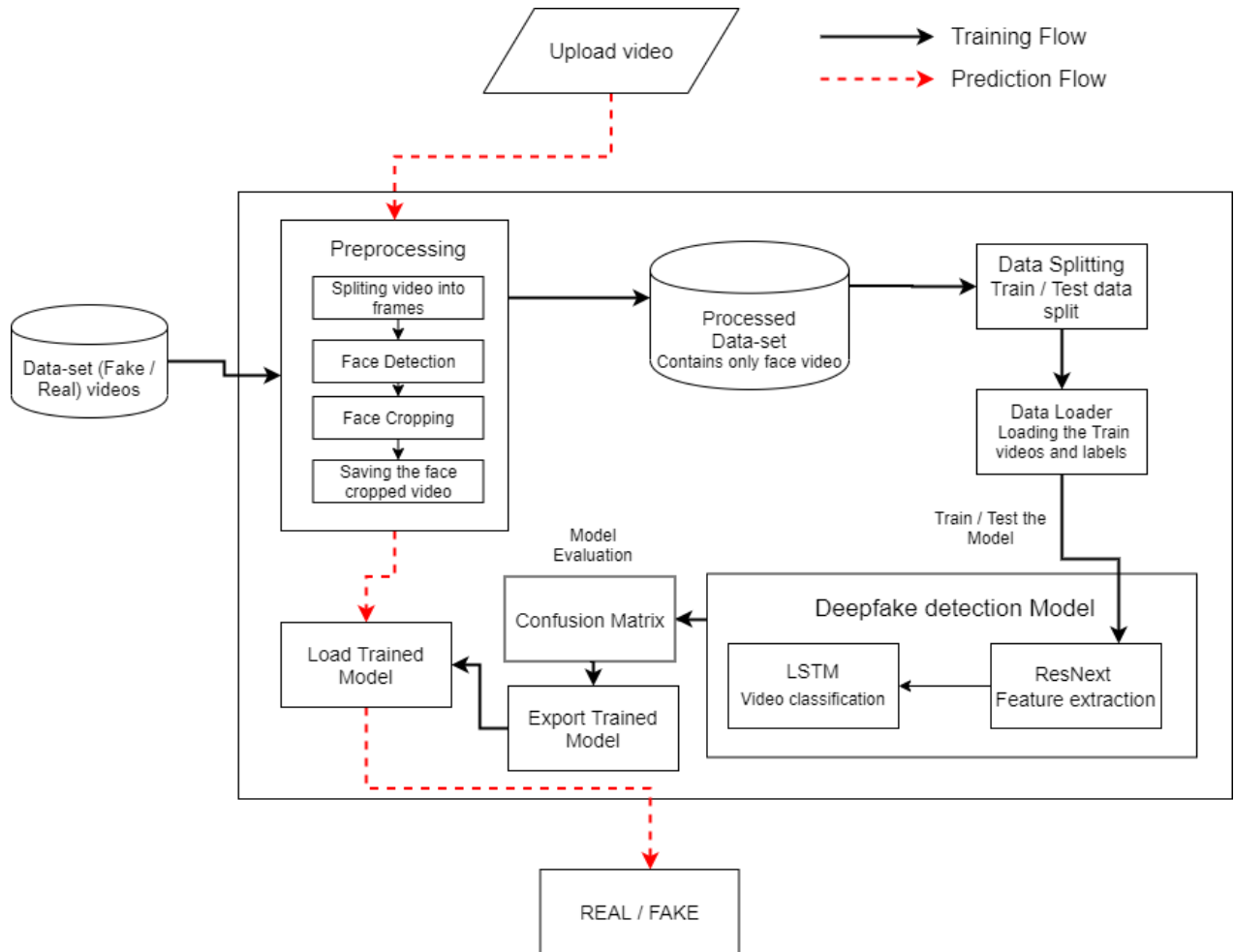


Fig:1 System Architecture

### 4.1 Introduction

The process commences with the upload of a video for analysis. Subsequently, the video undergoes preprocessing steps, which involve resizing the video dimensions, identifying and cropping facial regions within the video, and computing optical flow for compressed videos. These preprocessing measures are implemented to prepare the video data for subsequent analysis. The preprocessed data is then partitioned into training and testing datasets, each appropriately labeled to indicate whether the videos depict authentic content or deepfakes.

The training phase entails loading the labeled training data into a deep learning model for training purposes. The model architecture comprises two primary components: LSTM (Long Short-Term Memory) for video classification and ResNext for feature extraction. These components collaborate to discern the distinguishing patterns and characteristics of genuine

and deepfake videos. Throughout the training process, a confusion matrix is computed to assess the model's performance across both training and testing datasets.

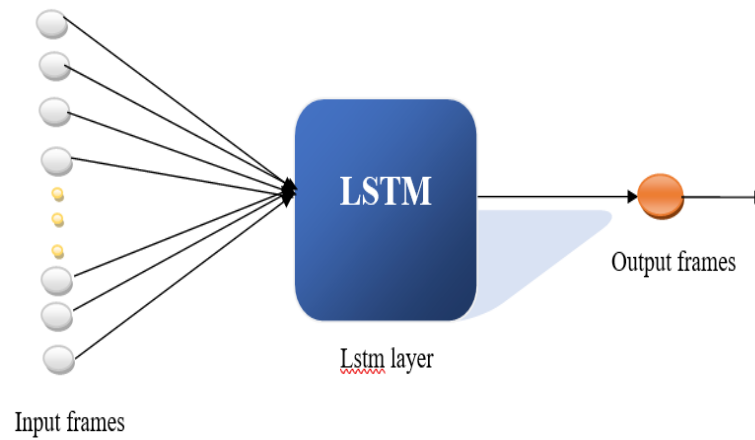


Fig 2: LSTM Architecture

This evaluation facilitates the determination of the model's accuracy in discriminating between authentic and deepfake videos. Upon completion of training and evaluation, the model is exported and poised for utilization in the prediction phase. During prediction, a new video is uploaded and subjected to the same preprocessing steps as employed in the training phase. The preprocessed video data, containing exclusively facial segments, is then inputted into the trained deepfake detection model. Leveraging the LSTM and ResNext components, the model scrutinizes the video content and issues a prediction regarding its authenticity – either REAL or FAKE. This prediction is grounded in the learned patterns and features gleaned throughout the training process. The outlined workflow encapsulates an end-to-end process for the development and deployment of a deep learning model proficient in detecting deepfake videos. It encompasses data preprocessing, model training, evaluation, and prediction, leveraging advanced techniques such as LSTM and ResNext to accurately classify videos as genuine or deepfakes.

## 4.2 Architectural Design

### 4.2.1 Module 1 : Dataset

To enhance the model's exposure to diverse deepfake techniques and improve generalizability, we leverage multiple datasets:

Face Forensics++ (ff++): This extensive dataset encompasses a variety of deepfakes created using different techniques, providing a solid base for training models.

Deepfake Detection Challenge (DFDC): This large-scale dataset includes both authentic and fake videos, though it may feature audio-manipulated content. Since our research is centered on visual deepfakes, we handle this by eliminating audio elements.

Celeb-DF: This dataset focuses on deepfakes involving celebrity and YouTuber faces, adding further variety to our training materials.

our model undergoes training with a wide variety of datasets, which include numerous methods of deepfake generation and various authentic situations. This exposure broadens the model's adaptability across different contexts. In addition, we enhance the dataset by integrating data augmentation strategies like arbitrary cropping, horizontal flips, and variations in color tones. Such methods serve to expand the training dataset's volume and variety, all while negating the need for gathering more video data.

|                     | FaceForensics++ [7]                                             | Celeb-DF [8]                                                            | DFDC [9]                                                                  |
|---------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Released            | Jan, 2019                                                       | Nov, 2019                                                               | Oct, 2019                                                                 |
| Tota no of videos   | 4000                                                            | 1203                                                                    | 5250                                                                      |
| Real videos         | 1000                                                            | 408                                                                     | 1131                                                                      |
| Fake videos         | 3000                                                            | 795                                                                     | 4119                                                                      |
| Resolution          | 480p, 720p, 1080p                                               | various                                                                 | 180p – 2160p                                                              |
| Format              | H.264, CRF=0, 23, 40                                            | MPEG4                                                                   | H.264                                                                     |
| Visual quality      | low                                                             | high                                                                    | high                                                                      |
| Temporal flickering | yes                                                             | improved                                                                | improved                                                                  |
| modality            | visual                                                          | visual                                                                  | Audio/visual                                                              |
| pros                | Poses a greater challenge due to the variety of deepfake types. | Offers straightforward access and primarily targets facial alterations. | Generally used for benchmarking, diverse deepfake creation techniques.    |
| cons                | Limited video length.                                           | May have limitations in applying to different forms of deepfakes.       | Few annotations exist for specific forms of deepfake manipulation videos. |

Table 3: Comparison of datasets

#### 4.2.2 Module 2 : Pre-processing

During this phase, the videos undergo preprocessing to eliminate any unwanted noise, focusing specifically on the facial area. Initially, we break down the video into frames. Following that, facial detection is carried out on each frame, trimming it down to include only the identified face. These trimmed frames are then compiled into a new video, creating a dataset consisting exclusively of face-centered videos. Frames lacking facial content are excluded during preprocessing to ensure dataset consistency. To effectively manage computational resources and maintain a consistent frame count, we establish a threshold value based on the mean total frame count per video. Recognizing computational limitations, especially concerning GPU capabilities, we opt for a threshold of 200 frames. This decision ensures manageable processing, considering that a 6.67-second video at 30 frames per second would yield 200 frames, avoiding computational overload. Consequently, only the first 200 frames of each video are preserved in the new dataset.

To demonstrate the efficacy of Long Short-Term Memory (LSTM), frames are analyzed sequentially rather than randomly. The resulting videos are saved at a frame rate of 30 frames per second.

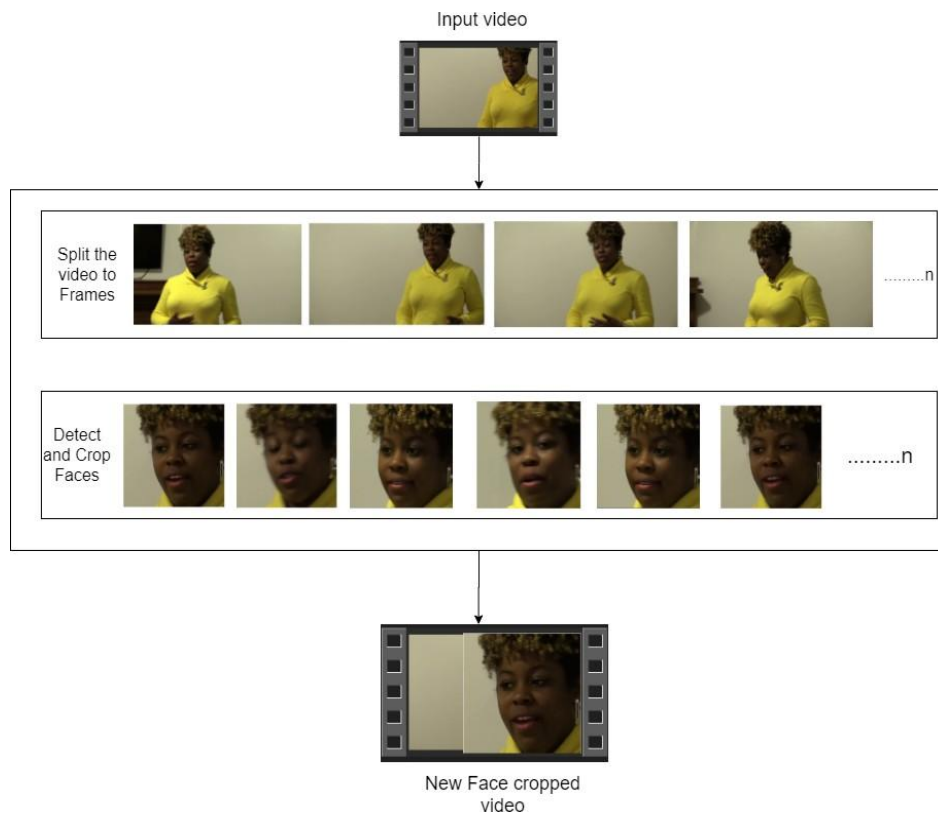


Fig 3: Pre-processing of video

#### 4.2.3 Module 3: Data-set split

We meticulously divided the dataset into two segments: training and testing of dataset. Allocating 80% of the videos for training and remaining 20% for testing ensures a balanced mix of real and fake videos in both sets. This approach is crucial for effectively training and evaluating our model, as it helps prevent overfitting and ensures the model's generalization capability. To enhance the robustness of our model, we incorporated techniques such as cross-validation and stratified sampling. Cross-validation allows us to assess the model's performance across multiple folds of the data, providing insights into its stability and reliability. Stratified sampling ensures that each class (real and fake videos) is represented proportionally in both the training and testing sets, minimizing the risk of biased performance evaluation. data augmentation strategies were implemented during the training phase to enrich the diversity of our training dataset. Methods such as image rotation, scaling, and mirroring were utilized during the preprocessing of the training images, aiding the model in learning invariant features and improving its ability to generalize to unseen data.

#### 4.2.4 Module 4: Model Architecture

Our proposed model architecture used the strengths of ResNeXt-50, for highly effective Convolutional Neural Network (CNN) for extracting feature of video , and Long Short-Term Memory (LSTM) networks are used for sequencing video frames by analyzing captured temporal dependencies. The preprocessed video which contains face only video frames are first fed into the ResNeXt-50 CNN, which extracts rich spatial features from each frame. The ResNeXt-50 architecture, renowned for its exceptional performance in image recognition tasks, ensures that even the most subtle facial details and inconsistencies are captured effectively. The extracted features from the ResNeXt-50 CNN are then passed into the LSTM component of our model. LSTMs, Recurrent Neural Networks (RNNs) are particularly adept at managing data that unfolds over time, which makes them well-suited for studying the evolution of facial expressions and movements across multiple frames. This capability allows them to track and interpret how expressions and movements change over time, offering valuable insights into the progression and context of these dynamic patterns.

#### **ResNext :**

ResNeXt architectures are crafted to enhance the capabilities of Residual Networks (ResNets) by integrating parallel convolution paths, referred to as "cardinality." This approach enables increased model capacity and refined feature representation. Take the ResNeXt-50 (32x4d)

architecture, for example, which employs 32 parallel convolution paths organized into four dimensions. In the initial conv1 stage, the input image undergoes processing through seven parallel 7x7 convolutions with a stride of 2, followed by a 3x3 max pooling layer with the same stride. Subsequent stages (conv2 to conv5) comprise blocks with diminishing input sizes and escalating numbers of parallel convolution paths. These paths empower the network to capture diverse and enriched features from the input data. Each block within these stages typically consists of 1x1, 3x3, and 1x1 convolutions, with the 3x3 convolutions serving as the parallel paths. For instance, a notation like 3x3, 128, C=32 signifies 32 parallel 3x3 convolutions operating on 128 input channels. Following the conv5 stage, the output undergoes global average pooling to amalgamate spatial information across feature maps. This pooled representation is then fed into a fully connected layer with 1000 units, succeeded by softmax activation for classification tasks.

| stage     | output  | ResNeXt-50 (32×4d)                                                                                      |
|-----------|---------|---------------------------------------------------------------------------------------------------------|
| conv1     | 112×112 | 7×7, 64, stride 2                                                                                       |
|           |         | 3×3 max pool, stride 2                                                                                  |
| conv2     | 56×56   | $\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128, C=32 \\ 1 \times 1, 256 \end{bmatrix} \times 3$    |
| conv3     | 28×28   | $\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256, C=32 \\ 1 \times 1, 512 \end{bmatrix} \times 4$    |
| conv4     | 14×14   | $\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512, C=32 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$   |
| conv5     | 7×7     | $\begin{bmatrix} 1 \times 1, 1024 \\ 3 \times 3, 1024, C=32 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$ |
|           | 1×1     | global average pool<br>1000-d fc, softmax                                                               |
| # params. |         | <b>25.0×10<sup>6</sup></b>                                                                              |

Fig 4: ResNext Architecture

## LSTM :

In our project for detecting deepfakes, we employ the Long Short-Term Memory (LSTM) architecture as a core component of our model, complemented by the integration of 2048-dimensional feature vectors. These vectors, extracted through techniques such as ResNeXt-50, encapsulate detailed spatial information within each frame of the video. Our methodology involves meticulously sequencing these 2048-dimensional feature vectors over time using LSTM. This process is akin to assembling individual pieces of a puzzle to unveil the broader

narrative of the video. Through this sequential analysis, our model discerns subtle temporal patterns indicative of deepfake manipulations, allowing us to effectively distinguish between authentic footage and manipulated content. Additionally, we leverage rectified linear units (ReLU) to introduce non-linearity into our model. Serving as the activation function within our neural network, ReLU aids in capturing intricate relationships and nuances present in the data. The combined utilization of LSTM, 2048-dimensional feature vectors, and ReLU empowers our model to extract insights from the temporal dynamics of video sequences, facilitating precise predictions regarding the authenticity of the content. By transcending the conventional analysis of individual frames and focusing on understanding the collective narrative they depict over time, our project represents an innovative approach to deepfake detection. Through the integration of LSTM for sequential analysis, 2048-dimensional feature vectors for spatial information, and ReLU for non-linear mapping, we provide a robust defense against digital deception.

#### 4.2.5 Module 5: Hyper-parameter tuning

To improve the model's performance and optimize the training process, we engage in hyperparameter tuning. This involves adjusting key parameters such as the learning rate (set at  $1 \times 10^{-5}$ ), optimizer choice (using Adam with a weight decay of  $1 \times 10^{-5}$ ), and dropout rate to achieve the best performance on a validation set. We employ strategies like grid search or random search to explore various hyperparameter combinations and find the configuration that delivers the best outcomes for our model. This approach allows us to fine-tune the model for exceptional performance in deepfake detection and ensures it generalizes effectively to new data. To use the available computation power properly the batch training is used. The batch size is taken of 4. Batch size of 4 is tested to be ideal size for training in our development environment.

### 4.3 Training Details

To effectively train the deepfake detection model, we divide the preprocess dataset into two parts: a training set and a test set, also referred to as a validation set. The training set comprises roughly 80% of the data. Its purpose is to train the model in distinguishing deepfakes. The remaining 20%, samples, constitute the test set, which plays a crucial role in evaluating the model's performance and its ability to generalize to unseen data. The training set is evenly balanced, consisting of 1,986 real samples and 1,842 fake samples, providing a diverse mix for the model to learn from. Similarly, the test set includes 495 real samples and 463 fake samples, serving as a benchmark for assessing the model's performance on data it hasn't encountered

during training. Throughout the training phase, the model learns from examples within the training set. Subsequently, the model undergoes evaluation on the test set to ascertain its capacity to generalize and mitigate overfitting, a phenomenon where the model becomes overly tailored to the specifics of the training data, led to underwhelming performance when applied to new datasets.

Efficient management of system resources during training is essential. Excessive worker processes for data loading can cause slowdowns or system hangs, necessitating the configuration of an appropriate number of worker processes based on the system's capabilities. Additionally, ensuring that the input data remains within valid ranges is vital to prevent problems like data clipping during the model's processing. This evaluation aids in assessing the model's proficiency in handling unseen data and mitigating overfitting, a scenario where the model excessively memorizes the training data, leading to inferior performance on new data.

#### 4.3.1 Early Stopping to Prevent Overfitting

Striving for an ideal equilibrium between a model's capacity to learn from training data and its capability to generalize to unseen instances stands as a crucial focus in machine learning. The visual representations, shown in Figure 5 and Figure 6, clearly demonstrate overfitting, which occurs when the model becomes overly adapted to the training data, leading to diminished performance on new samples. While the training loss curve in Image 1 exhibits a steady descent, indicating proficiency in fitting the training data, the validation loss curve displays an erratic nature, ultimately escalating after a certain point. Corroborating this observation, the accuracy curves in Image 5 reveal a stark contrast – the training accuracy soars, yet the validation accuracy plateaus, underscoring the model's limited generalization capabilities. To circumvent this predicament, the judicious implementation of an early stopping technique proves indispensable. This approach entails monitoring the validation loss during the training process, as shown in fig 5, and terminating the training When the validation loss stops improving after a set number of epochs, early stopping is implemented. This strategy halts the training process at the right time, ensuring that the model's ability to generalize is preserved while making the most of its learned representations. By doing so, overfitting is mitigated, and the model's performance on unseen data is enhanced. This can be observed in the stable validation accuracy depicted in Figure 6.

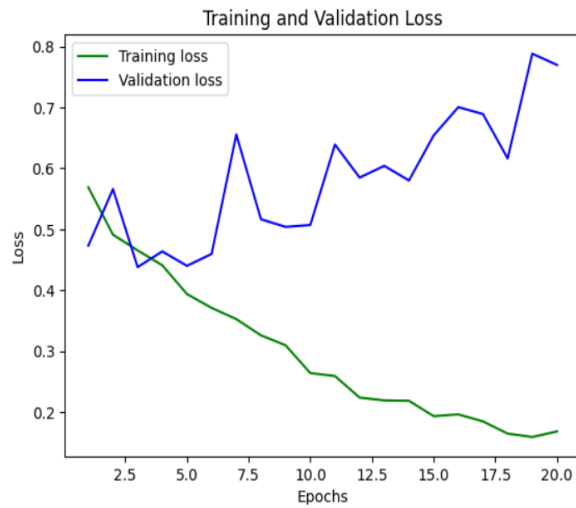


Fig 5: validation loss

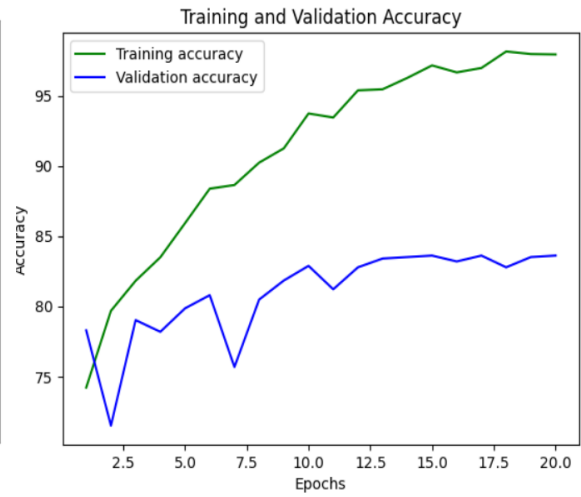


Fig 6: validation accuracy

#### 4.4 Model Prediction Details

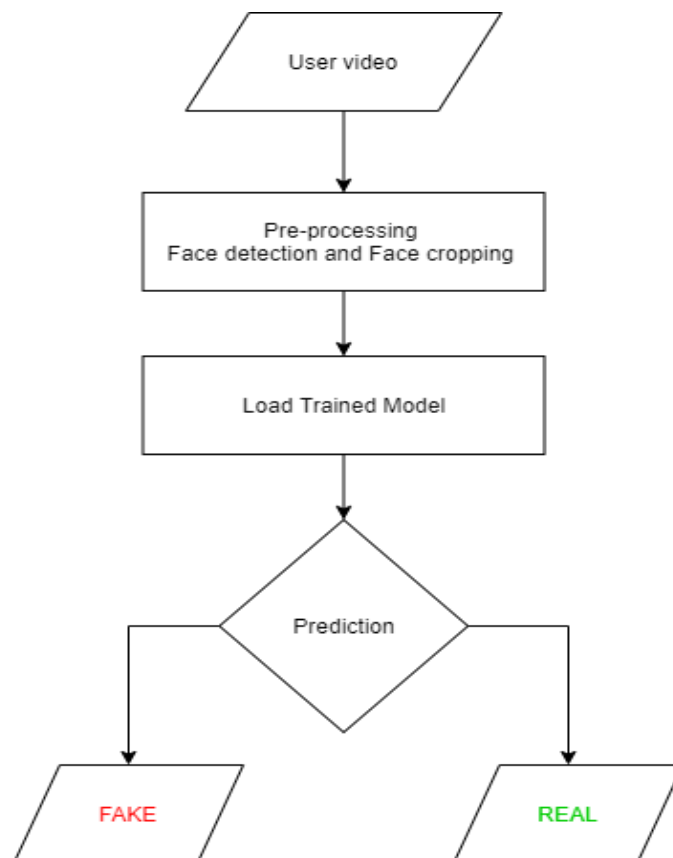


Fig 7: Testing workflow

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The process begins when a user uploads or provides a video for analysis. The video undergoes preprocessing steps, including face detection and face cropping techniques. These steps identify and extract the facial regions from the video frames, as facial data is crucial for accurate deepfake detection. A pre-trained deep learning model, specialized in distinguishing real videos from fake (deepfake) ones, is then loaded into the system. This model has been extensively trained on a massive dataset comprising both authentic and artificially manipulated (deepfake) videos, allowing it to learn the intricate patterns and features that differentiate between the two. The preprocessed facial data extracted from the user's video is fed into this loaded trained model. The model meticulously analyzes the data, leveraging its knowledge gained during the training process, and makes a prediction on whether the input video is REAL or FAKE (deepfake). Based on this prediction, the system outputs the corresponding result. If classified as FAKE, it indicates that the model has identified the user's video as an artificially manipulated or synthesized deepfake. Conversely, if classified as REAL, it signifies that the video is considered an authentic representation of the subject, free from any deepfake manipulations.

After outputting the prediction, additional actions or processes may be triggered based on the result. If a deepfake is detected, the system could implement countermeasures like flagging the video for review, blocking its distribution, or notifying relevant authorities. For authentic videos, subsequent processing or analysis steps could be executed as per the specific application requirements. The system may also incorporate feedback mechanisms to continuously improve the deepfake detection model's accuracy and robustness, enabling it to adapt to new deepfake techniques or edge cases. Ongoing research aims to enhance such systems' capabilities in combating the ever-evolving deepfake technology landscape.

## CHAPTER 5: RESULT

In the implementation of model, we perform prediction on real and fake of same person then we found that model is able to predict whether video is real or fake as we shown on the fig (5, 6). fig 5 shows a real video with 95% confidence accuracy which is correct as we perform on real video but when we perform fake video of same person in previous video then model able to predict the video is fake with the 96% confidence. all over accuracy of model is 94%. Hence our model is able to predict whether the video is manipulated or not, showing in the results video is real or fake.

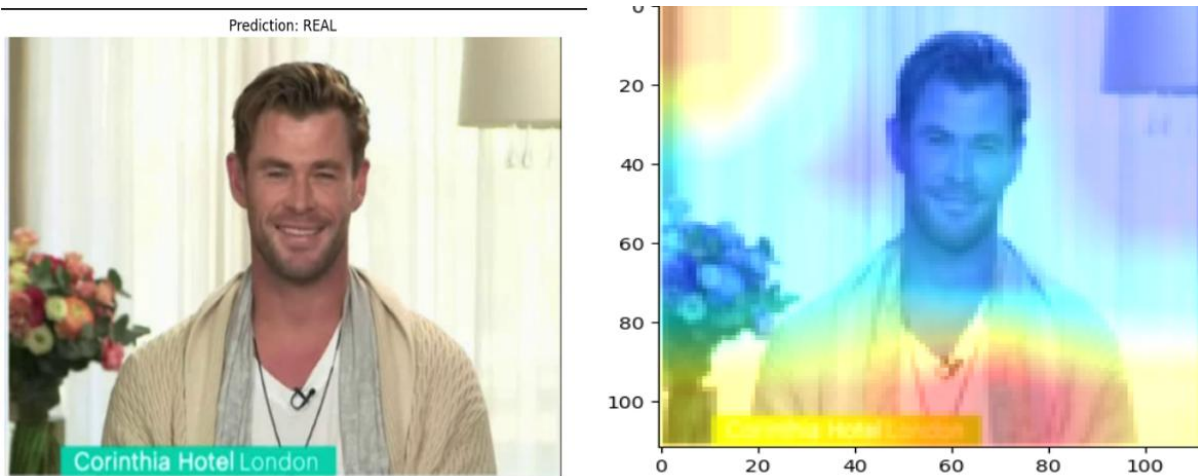


Fig 8 : prediction on real video

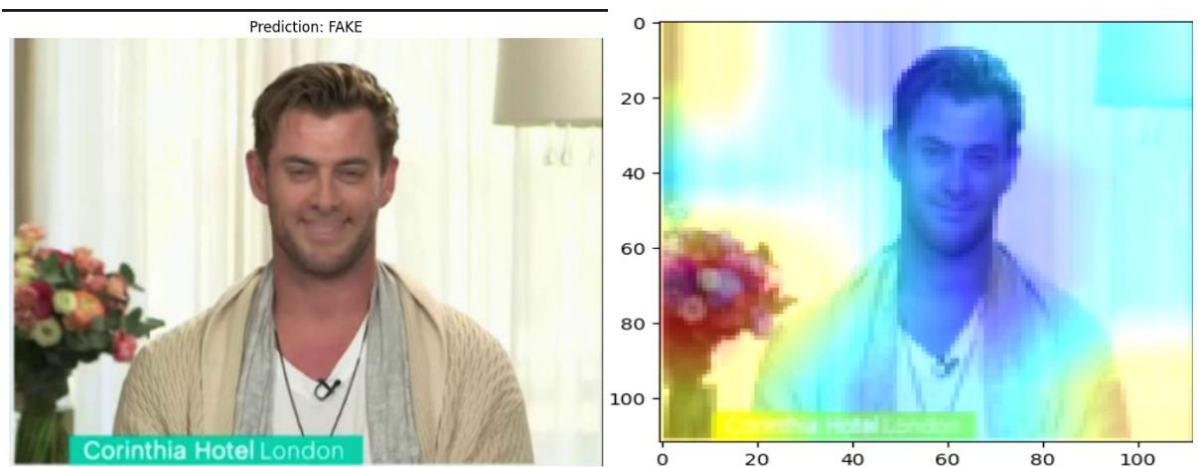


Fig 9 : prediction on deepfake video

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## CHAPTER 6: CONCLUSION & FUTUERE SCOPE

### 6.1 Conclusion

This paper introduces an advanced approach utilizing neural networks to classify videos as either real or deepfake, while also showcasing the confidence level of the model's predictions. The approach was developed given that there are many methods of creating such videos, the mentioned was implemented using GANs and autoencoders. The main goal is to check how often the video detection using the identified modifications can be correct and the video – fake or real using the discussed parameter can be correct. This method can provide high accuracy, especially when working in real time. we propose a neural networkbased method of classifying videos into real and deepfake, while yielding the confidence level of the model predictions. Our approach utilizes a ResNext50 architecture for detecting frame-level features and employs an LSTM with 2048 dimensions. The objective of our project is to verify the success of the video detection based on modifications and classify all the videos to either real or fake according to our stated parameters. We present a new method of classifying videos into deepfake or real, along with the provided confidence level of model predictions. Since the deepfake generation algorithm uses autoencoders and GAN, our method uses ResNext CNN for frame detection and an RNN network with LSTM to classify the video.

### 6.2 Future Scope

Our approach demonstrates the capability to classify videos as either real or deepfake based on the parameters outlined in the paper. We have used ResNext-50 with LSTM for face-only videos, but the model can be further developed in the future to detect any type of deepfake, including alterations to any part of the body shown in the video. Additionally, the model could be enhanced to detect malicious audio in a video with high accuracy by integrating it with different models. In the future, a multimodal approach could be employed to achieve greater deepfake detection accuracy. This approach would involve combining the strengths of various models like Attention Mechanisms, VGG, Inception, Efficient Net and other neural network architectures (e.g., transformers Vision Transformer (ViT) and capsule networks). Such a comprehensive system would be capable of detecting any type of deepfake in videos, images, and audio, improving overall performance and robustness.

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